

## PRACTICAL 7

### PRACTICAL TITLE :- PHP Form Processing with Server-Side Validation and CSV/JSON File Storage

#### Practical 7: PHP Form Processing with Server-Side Validation and CSV/JSON File Storage

|                                                  |                                                                                                                                                                                           |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Problem Definition</b>                        | Process submitted registration/contact form data using PHP. Validate and sanitize inputs on the server side and store records in CSV or JSON file format. Display success/error messages. |
| <b>Key Questions / Analysis / Interpretation</b> | 1. Is the form submitted using POST?<br>2. Are inputs validated and sanitized server side?<br>3. Is file writing handled safely?<br>4. Are success and error responses displayed clearly? |
| <b>Key Skills to be Addressed</b>                | PHP forms, POST handling, server-side validation, sanitization, file writing, CSV/JSON storage                                                                                            |
| <b>Applications</b>                              | Backend form processors and small data collection systems                                                                                                                                 |
| <b>Learning Outcome</b>                          | Students will implement server-side form processing with safe file-based storage.                                                                                                         |
| <b>CO Mapping</b>                                | CO1, CO5                                                                                                                                                                                  |
| <b>Dataset/Test Data</b>                         | Registration/contact form inputs                                                                                                                                                          |
| <b>Tools/Technology to be Used</b>               | PHP, XAMPP/WAMP/LAMP, browser dev tools                                                                                                                                                   |
| <b>Total Hours</b>                               | Implementation: 4 hours   Total engagement: 6 hours                                                                                                                                       |
| <b>Post Laboratory Work Description</b>          | Submit PHP source code, test cases, and generated CSV/JSON file.                                                                                                                          |
| <b>Evaluation Strategy Including Viva</b>        | POST handling, validation, sanitization, storage correctness, and viva.                                                                                                                   |
| <b>Advanced/Intermediate Extension</b>           | Intermediate: Display stored CSV/JSON records on a webpage.<br>Advanced: Add CSRF token validation for form submission.                                                                   |

### GIT HUB REPOSITORY :-

<https://github.com/nehvala0005-cloud/sh.git>

#### 1. Aim

To develop a StudentHub registration form using PHP for server-side form processing, validation and sanitization, and to store the submitted student records in a CSV file.

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## **2. Problem Definition**

**Process the StudentHub registration form data using PHP. The submitted data is sent through the POST method, validated and sanitized on the server side, and valid student records are stored in a CSV file. Appropriate success and error messages are displayed according to the form submission result.**

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## **3. Key Questions / Analysis / Interpretation**

- 1. Is the registration form submitted using the POST method?**
  - 2. Are the submitted inputs validated and sanitized on the server side?**
  - 3. Is the student data stored correctly in the CSV file?**
  - 4. Are success and error responses displayed clearly?**
  - 5. Is invalid data prevented from being stored?**
- 

## **4. Key Skills to be Addressed**

- PHP form processing**
  - POST request handling**
  - Server-side validation**
  - Input sanitization**
  - CSV file handling**
  - File writing**
  - Error handling**
  - HTML forms**
  - JavaScript form validation**
  - Client-server communication**
- 

## **5. Applications**

- **Student registration systems**
  - **Contact form processors**
  - **Student data collection systems**
  - **Small backend data-management systems**
  - **File-based registration systems**
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## **6. Theory**

**PHP is a server-side scripting language used to process data submitted through web forms.**

**In this practical, the StudentHub registration form collects student information such as:**

- **Name**
- **Email**
- **Mobile number**
- **Date of Birth**
- **Gender**
- **Course**
- **Address**

**When the user submits the form, JavaScript performs initial validation and sends the form data to the PHP file using a POST request.**

**The PHP script then:**

- 1. Receives the submitted data.**
- 2. Checks whether the request uses POST.**
- 3. Validates the submitted information.**
- 4. Sanitizes the input data.**
- 5. Opens/creates the CSV file.**
- 6. Stores the valid student record.**

7. Returns a success or error response.

The CSV file provides simple file-based storage without requiring a database.

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## 7. Technologies / Tools Used

| Tool             | Purpose                    |
|------------------|----------------------------|
| HTML5            | Creating registration form |
| CSS3             | Styling the form           |
| JavaScript ES6   | Client-side validation     |
| PHP              | Server-side processing     |
| XAMPP            | Local PHP server           |
| CSV              | Storing student records    |
| Browser DevTools | Testing and debugging      |
| VS Code          | Writing source code        |

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## 8. Dataset / Test Data

The StudentHub registration form uses student registration details as test data.

Example:

| Field         | Sample Value                                                                  |
|---------------|-------------------------------------------------------------------------------|
| Name          | Nehal Vala                                                                    |
| Email         | <u><a href="mailto:nikhilvala0005@gmail.com">nikhilvala0005@gmail.com</a></u> |
| Mobile        | 9624480557                                                                    |
| Date of Birth | 08-12-2007                                                                    |
| Gender        | Male                                                                          |

| Field   | Sample Value    |
|---------|-----------------|
| Course  | B.Tech CE       |
| Address | Student address |

Invalid test data was also used, such as:

Mobile: 12345

Expected result:

Enter exactly 10 digits.

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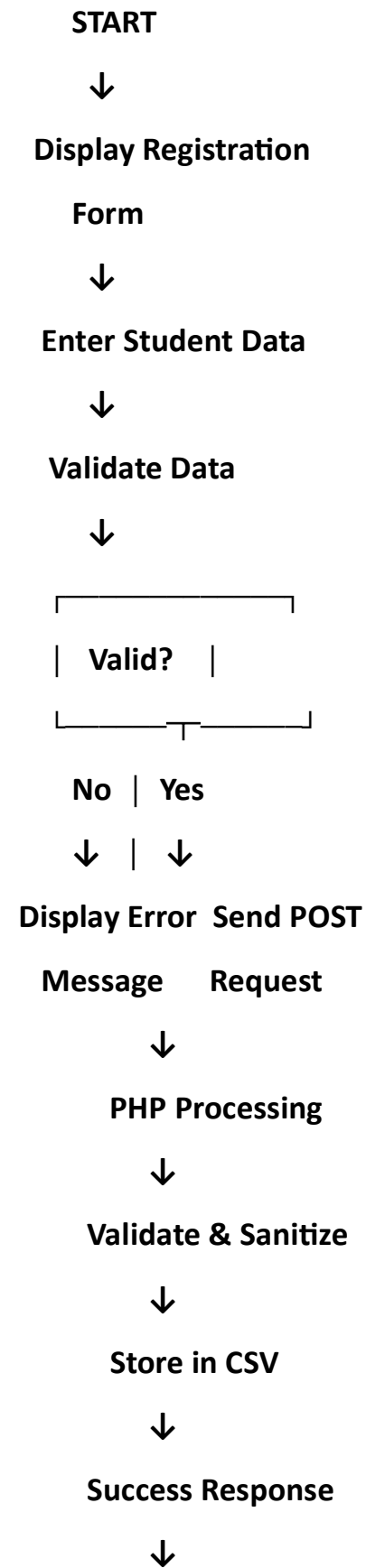
## 9. Algorithm

### Algorithm for StudentHub Registration

1. Start.
  2. Display the StudentHub registration form.
  3. Enter student details.
  4. Validate the entered data.
  5. If the data is invalid, display an appropriate error message.
  6. If the data is valid, create a FormData object.
  7. Send the form data to the PHP file using the POST method.
  8. PHP receives the submitted data.
  9. Validate and sanitize the received data.
  10. Open the CSV file.
  11. Write the student record into the CSV file.
  12. Return a success response.
  13. Display the registration success popup.
  14. Stop.
- 

## 10. Flowchart

You can draw this simple flowchart in your practical file:



**Success Popup**



**END**

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## **11. Implementation**

**The StudentHub registration system consists of:**

**Frontend**

**register.html**

**register.css**

**register.js**

**Backend**

**register.php**

**Storage**

**students.csv**

**The registration form submits data to the PHP backend. PHP processes the submitted information and stores valid records in students.csv.**

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## **12. PHP Processing**

**The PHP backend performs the following operations:**

- **Checks the request method.**
- **Receives POST data.**
- **Validates required fields.**
- **Sanitizes user input.**
- **Opens the CSV file.**
- **Writes the student record.**
- **Returns a JSON response.**

**The generated CSV file contains fields such as:**

**Name**

**Email**

**Mobile**

**Date of Birth**

**Gender**

**Course**

**Address**

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### **13. Validation Testing**

#### **Test Case 1 — Valid Registration**

**Input:** All fields contain valid data.

**Expected Result:**

Registration is successful and the data is stored in the CSV file.

**Actual Result:**

Registration successful.

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#### **Test Case 2 — Invalid Mobile Number**

**Input:**

**12345**

**Expected Result:**

Enter exactly 10 digits.

**Actual Result:**

Validation error displayed and form submission is stopped.

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#### **Test Case 3 — Invalid Email**

**Input:**

**nehal@**

**Expected Result:**

**Enter a valid email address.**

**Actual Result:**

**Invalid email is rejected.**

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#### **Test Case 4 — Weak Password**

**Input: Password that does not satisfy the required conditions.**

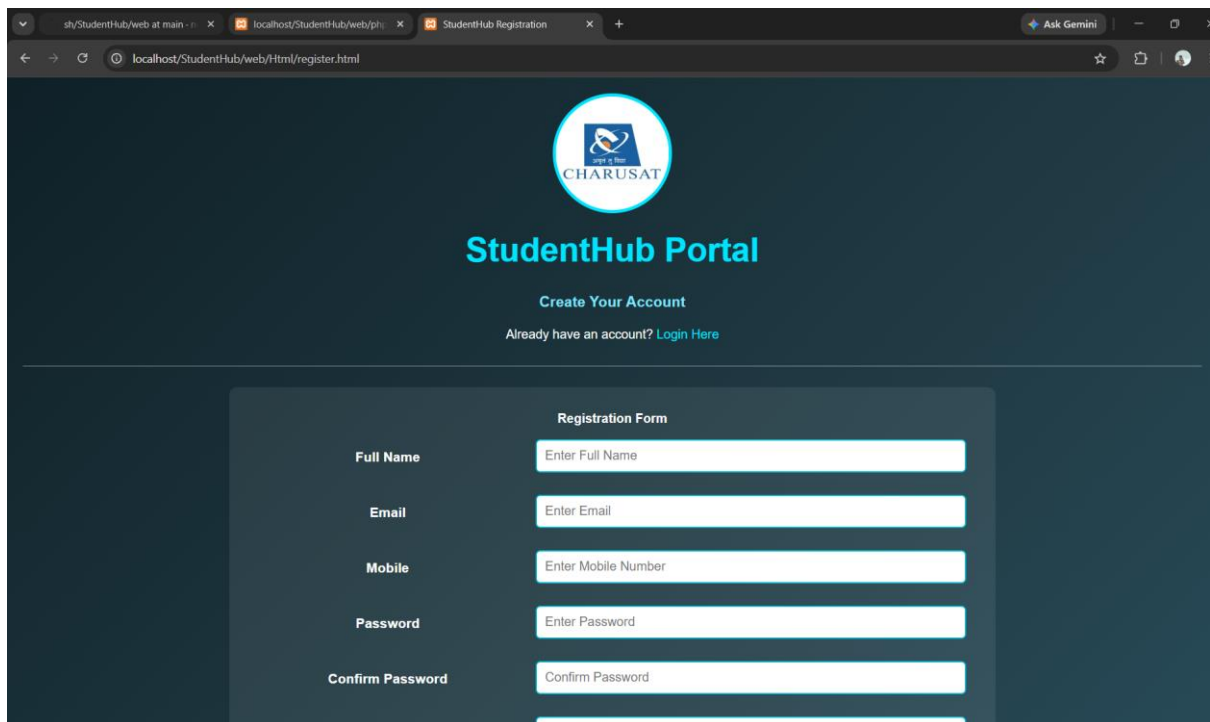
**Expected Result:**

**Use uppercase, lowercase and a number.**

**Actual Result:**

**Password validation message is displayed.**

## **14 . PROOF OF WORK :-**

A screenshot of a web browser displaying the StudentHub Portal registration form. The browser's address bar shows the URL 'localhost/StudentHub/web/Html/register.html'. The page features a dark blue header with the CHARUSAT logo and the text 'StudentHub Portal'. Below the header, there is a 'Create Your Account' section with a link to 'Login Here' for existing users. The main content area contains a 'Registration Form' with five input fields: 'Full Name', 'Email', 'Mobile', 'Password', and 'Confirm Password'. Each field has a placeholder text indicating what to enter (e.g., 'Enter Full Name', 'Enter Email', etc.).

**Fig. 1: StudentHub registration form for collecting student information.**

The screenshot shows a web browser at the URL `localhost/StudentHub/web/html/register.html`. The page features the CHARUSAT logo and the title "StudentHub Portal". Below the title, there is a "Create Your Account" section with a link "Already have an account? Login Here". The registration form is titled "Registration Form" and contains the following fields:

- Full Name: Nehal Vala
- Email: nikhilvala0005@gmail.com
- Mobile: 123456 (with a red border and a red error message "Enter exactly 10 digits.")
- Password: (masked with asterisks, with a green message "Strong password")
- Confirm Password: (masked with asterisks)
- Date of Birth: 02-09-2026
- Gender: ☒ Male ☐ Female ☐ Other
- Course: B.Tech CE

**Fig. 2: Validation error displayed for an invalid mobile number.**

The screenshot shows the same web browser at the URL `localhost/StudentHub/web/html/register.html`. The page features the CHARUSAT logo and the title "StudentHub Portal". Below the title, there is a "Create Your Account" section with a link "Already have an account? Login Here". The registration form is titled "Registration Form" and contains the following fields:

- Full Name: Nehal Vala
- Email: nikhilvala0005@gmail.com
- Mobile: 123456
- Password: (masked with asterisks, with a green message "Strong password")
- Confirm Password: (masked with asterisks)
- Date of Birth: 12-09-2026
- Gender: ☒ Male ☐ Female ☐ Other
- Course: B.Tech CE

A modal dialog box is displayed in the center of the form, containing a green checkmark icon, the text "Congratulations!", the message "You are registered successfully.", and a green button labeled "Continue to Login".

**Fig. 3: Successful registration message displayed after form processing.**

|   | A          | B            | C         | D            | E      | F         | G                                                                                                                 | H | I | J | K | L | M | N | O | P | Q | R | S | T | U |
|---|------------|--------------|-----------|--------------|--------|-----------|-------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Name       | Email        | Mobile    | Date of Birt | Gender | Course    | Address                                                                                                           |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 2 | Nehal Vala | nikhilvala00 | 9.624E+09 | 08-12-2007   | on     | B.Tech CE | 202 ,<br>ashirwad<br>hostel NR<br>Charusat<br>University<br>Main gate ,<br>valetva<br>road ,<br>changa ,<br>anand |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 3 | Nehal Vala | nehal@exar   | 9.624E+09 | 13-09-2026   | on     | B.Tech CE | 202 ,<br>ashirwad<br>hostel NR<br>Charusat<br>University<br>Main gate ,<br>valetva<br>road ,<br>changa ,<br>anand |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 4 | Nehal Vala | nikhilvala00 | 1.235E+09 | 03-09-2026   | on     | B.Tech CE | 202 ashirwad hostel Nr Charusat University main gate, valetva road , changa , anand, 388421                       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 5 | Nehal Vala | nikhilvala00 | 9.624E+09 | 12-09-2026   | on     | B.Tech CE | 202 ,<br>ashirwad<br>hostel NR<br>Charusat<br>University<br>Main gate ,<br>valetva<br>road ,<br>changa ,<br>anand |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 6 |            |              |           |              |        |           |                                                                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 7 |            |              |           |              |        |           |                                                                                                                   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

**Fig. 4: Successfully stored student registration records in CSV file.**

## 15. Post Laboratory Work

- Submit the PHP source code.
- Submit the registration form source files.
- Submit validation test cases.
- Submit screenshots of valid and invalid inputs.
- Submit the generated students.csv file.
- Demonstrate successful server-side form processing.

## 16. Evaluation Strategy

The practical can be evaluated based on:

- Correct POST request handling
- Server-side validation
- Input sanitization
- Correct CSV file writing
- Proper error handling

- **Successful registration response**
  - **Correct storage of submitted records**
  - **Code walkthrough and viva**
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## **17. Intermediate / Advanced Extension**

### **Intermediate**

**Display the stored CSV records on a webpage using PHP.**

### **Advanced**

**Add CSRF token validation to protect the registration form from unauthorized submissions.**

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## **18. Learning Outcomes**

**Students will be able to implement server-side form processing using PHP, validate and sanitize submitted form data, handle POST requests, and store valid student records safely in CSV-based storage.**

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## **19. Result**

**The StudentHub registration form was successfully integrated with PHP for server-side processing. Valid student information was successfully validated and stored in the CSV file, while invalid inputs were rejected with appropriate error messages.**

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## **20. Conclusion**

**Thus, the StudentHub registration system was successfully implemented using PHP, HTML, CSS and JavaScript. Server-side form processing, validation, sanitization and CSV file storage were successfully demonstrated. The practical helped in understanding how frontend forms communicate with a PHP backend and how submitted data can be validated and stored using file-based storage.**

**Faculty Remarks :-** \_\_\_\_\_